Module 3 Summative Assessment

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

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Assignment details

Due
Nov 10, 11:59 PM IST

Attempts
0 left (1 attempts every 8 hours)

Try again

Submitted
Nov 16, 12:27 AM IST

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To pass you need at least 80%. We keep your highest score.

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Instructions

1. Why might you use ordinal encoding instead of one-hot encoding in a linear regression model? **1 / 1 point**

- [ ] To increase the computational efficiency of the model.
- [x] To preserve the order of the categories in a categorical variable.
- [ ] To handle datasets with a large number of missing values.
- [ ] To ensure that the model is highly interpretable.

Correct
Correct!

2. How does LDA differ from Principal Component Analysis (PCA)? **1 / 1 point**

- [ ] LDA is used for regression, while PCA is used for classification.
- [ ] LDA cannot reduce dimensionality, unlike PCA.
- [ ] LDA maximizes the variance within each class, while PCA maximizes the total variance.
- [x] LDA is a supervised method, whereas PCA is unsupervised.

Correct
Correct!

3. What type of variable is typically the target in logistic regression? **1 / 1 point**

- [x] A binary variable representing two classes or categories.
- [ ] A nominal variable with more than two categories.
- [ ] A continuous variable representing any real number.
- [ ] An ordinal variable with a natural ordering.

Correct
Correct! Logistic regression is often used for binary classification problems.

4. What is a residual in the context of linear regression? **1 / 1 point**

- [x] The difference between an observed value and its corresponding predicted value based on the regression model.
- [ ] The proportion of variance explained by the model.
- [ ] The gradient of the regression line.
- [ ] The average of all the dependent variable values.

Correct
Correct!

5. In LDA, what happens if the assumption of equal covariance matrices across different classes is violated? **1 / 1 point**

- [ ] The method becomes equivalent to PCA.
- [x] The performance of LDA may degrade, potentially leading to less accurate classifications.
- [ ] The number of discriminant functions increases.
- [ ] LDA will always fail to classify any observations correctly.

Correct
Correct!

6. What is one advantage of using logistic regression? **1 / 1 point**

- [x] It provides the probability of class membership, offering more information than just a classification.
- [ ] It can predict continuous values without any transformation.
- [ ] It requires no assumptions about the distributions of classes in feature space.
- [ ] It can handle highly correlated predictor variables without any issue.

Correct
Correct! This probability aspect is a key advantage of logistic regression.